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Improved chronologic resolution of the Hawthorn and the Alum Bluff Groups in northern Florida: Implications for Miocene chronostratigraphy

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ABSTRACT

Several geochronologic methods are used to constrain the ages of the early to middle Miocene Hawthorn and Alum Bluff Groups in the eastern Florida panhandle. The Dogtown Member of the upper Torreya Formation (Hawthorn Group) in northern Gadsden County contains an early Barstovian land-mammal fauna, has $^{87}Sr/^{86}Sr$ age estimates between 14.7 ± 1.5 and 16.6 ± 1.0 Ma, is of reversed magnetic polarity, and probably correlates with Chron C5B-R. These results constrain the age of the Dogtown Member in the study area to between about 15.3 and 15.9 Ma, significantly younger than previously recognized for the upper Torreya Formation. A second fossil locality from the Dogtown Member occurs in the same lithostratigraphic interval but is older, based on vertebrate biochronology. This indicates that the Dogtown Member is time transgressive, that is, younger to the north. Early Hemingfordian land-mammal faunas and an invertebrate fauna correlative with planktonic foraminiferal zones upper N5 and N6 are known from the lower Torreya Formation. One mollusc sample from the Seaboard locality produced a $^{87}Sr/^{86}Sr$ age estimate of 18.4 ± 1.0 Ma, consistent with the biochronology. The revised age of the Torreya Formation is late early to early middle Miocene, between about 19 and 15.3 Ma.

The Chipola Formation has been interpreted to be late early or middle Miocene (N7 and N8) in age and younger than the Torreya Formation, based on the superposition of the Chipola over the Torreya at the only known stratigraphic contact and apparent biochronologic differences. Four samples from the Chipola Formation at Alum Bluff yield $^{87}Sr/^{86}Sr$ age estimates between 18.3 and 18.9 (±1.0) Ma (early Miocene), much older than previous interpretations. A small land-mammal fauna from an overlying unit supports the older age for the Chipola Formation. The discrepancy in ages is probably due to poor biostratigraphic correlation rather than erroneous $^{87}Sr/^{86}Sr$ ages. The lithostratigraphic relationship between the Chipola and the Torreya Formations is poorly known from cores only; chronostratigraphic evidence indicates that the Chipola and Torreya Formations overlap in age and may interdigitate, possibly representing transitional marine and nearshore/shelf facies.

INTRODUCTION

The coastal plain of the southeastern United States is blanketed by extensive nearshore marine and transitional marine/nonmarine deposits of Tertiary age such as the Hawthorn and Alum Bluff Groups of northern Florida. The Hawthorn and Alum Bluff Groups are heterogeneous accumulations of predominantly Miocene clastic sediments, frequently phosphatic and fossiliferous. Individual formations and members of the Hawthorn and Alum Bluff Groups are lithologically variable, interdigitate, differ in age between outcrops, and are poorly exposed; traditional bio- and lithostratigraphic correlations have proven difficult. The confusing stratigraphic nomenclature of the coastal plain compounds the problem; a number of faunizones, magnafacies, stages, lithofacies, and other terms have been utilized for the same rock-stratigraphic units. Many units were defined and recognized by their faunal content, so that most were initially biostratigraphic units with rock-stratigraphic names (Waller, 1969; Schmidt, 1984). Recent lithostratigraphic efforts have done much to improve our knowledge of coastal-plain stratigraphy, particularly of the Hawthorn Group (Huddlestun, 1988; Scott, 1988). The relationships between the Hawthorn Group and other lithostratigraphic groups (such as the Alum Bluff Group) remain poorly resolved, however, because of a lack of chrono- and lithostratigraphic control.

Two basic views exist regarding the chronostratigraphic relationships between the Alum Bluff and Hawthorn Groups (Waller, 1969). The first, based primarily upon biostratigraphy, contends that many of the "formations" of the Alum Bluff and Hawthorn Groups represent different (that is, noncontemporaneous) intervals of time on the basis of their faunal content (Gardner, 1926; Cooke, 1945; Banks and Hunter, 1973). A second view, based primarily upon lithostratigraphy, contends that these units should be defined as lithostratigraphic units and that they basically represent different facies that are approximately contemporaneous (Puri and Vernon, 1964; Huddlestun, 1988; Scott, 1988). Unfortunately, the Hawthorn and Alum Bluff Groups have relatively few biochronologically useful fossils in common; thus, a fundamental problem in deciphering stratigraphic relationships has been a lack of direct chronologic control.

In this regard, strontium-isotopic ($^{87}Sr/^{86}Sr$) ratios have received considerable attention recently for correlation and dating of marine sequences across major environmental boundaries (see Koepnick and others, 1988, for a recent review). Strontium isotopes may provide independent, high-resolution age estimates for these deposits during certain periods of geologic time (for example, much of the Miocene Epoch) and have proved

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very useful in the Atlantic and Gulf coastal plains (for example, Webb and others, 1989). Nearshore marine sediments containing abundant invertebrate and vertebrate fossils are extensively exposed on the coastal plain, but until recently have been dated only by poorly constrained correlations to standard biochronologic time scales. Multiple, indirect correlations, the general lack of diagnostic planktonic foraminifera, and facies changes have led to confusion over the ages and correlations. Recent discoveries of fossil vertebrates at important localities and advances in $^{87}Sr/^{86}Sr$ chronostratigraphy have allowed an integration of several geochronologic methods to produce more refined age estimates than previously available for the Alum Bluff and Hawthorn Groups.

The purpose of this study is to present an integrated bio-, $^{87}Sr/^{86}Sr$, and magnetic polarity stratigraphy for the Torreya Formation of the Hawthorn Group and the Chipola Formation of the Alum Bluff Group, and use these data to propose revisions of chronostratigraphic relationships for these units in the Florida panhandle. This in turn enhances our understanding of stratigraphic relationships, sedimentation and sea-level cycles, vertebrate and invertebrate evolution and biostratigraphy, and biogeography of marine and terrestrial organisms preserved in the Alum Bluff and Hawthorn Groups.

LITHOSTRATIGRAPHY

Torreya Formation

The Torreya Formation is characterized by an overall siliciclastic composition, with increasing carbonate content toward the base (see Huddlestun, 1988, and Scott, 1988, for detailed discussions of lithology and nomenclatural history). The Dogtown Member is recognized for a clay-rich interval at or near the top of the Torreya Formation (Huddlestun and Hunter, 1982). Our field and laboratory investigations of the Dogtown Member focused on four open-pit clay mines: the Engelhard La Camelia Mine (stratotype of the Dogtown Member), the Milwhite Gunn Farm Mine, the Floridin Corry Mine, and the Floridin Smith Mine (Fig. 1). Each of these mines exposes about 15 m of the Dogtown Member, unconformably overlain by the late Miocene or Pliocene Citronelle and/or Miccosukee Formations (Fig. 2). The base of the Dogtown Member is not exposed. The Citronelle and Miccosukee Formations are predominantly heavily weathered, unfossiliferous sands, and were not included in the study. The Dogtown Member in the study area generally consists of one or two beds of fuller's earth clay (composed of palygorskite, montmorillonite, and sepiolite), overlain by about 12 m of mixed sand and clay with variable amounts of silt and carbonate (as calcite and dolomite cements). Invertebrate fossils, primarily molluscs, are found in virtually every bed within the Dogtown Member in the mines, with the exception of the mined fuller's earth clay beds.

Weaver and Beck (1977) proposed that the Dogtown Member, as represented in the present study area, records two cycles of sedimentation, preserved at or very near the ancient shoreline. An overall schizohaline, lagoonal or tidal environment is inferred from modern analogues of palygorskite, montmorillonite, and sepiolite deposition with penecontemporaneous dolomite formation (Weaver and Beck, 1977). Periods of subaerial exposure are indicated by pedogenic features such as mudcracks, root casts, hardgrounds, and organic-rich horizons (Patterson, 1974; Weaver and Beck, 1977; Kirk, 1983). Paleontological evidence for brackish or variable conditions includes the abundance of the benthic Foraminifera *Elphidium* sp. and *Ammonia beccarii* (Olson, 1966) and a low diversity of fossil marine molluscs (Gardner, 1926; Hunter and Huddlestun, 1982). Deposition of the Torreya Formation in close proximity to land is also indicated by the presence of terrestrial and fresh-water fossil vertebrates throughout the formation (Olsen, 1964b, 1968).

The Seaboard locality in Tallahassee is stratigraphically near the base of the Torreya Formation (Hunter and Huddlestun, 1982). The outcrop today is heavily overgrown, but *in situ* molluscs were recovered from the base of the outcrop, presumably from unit 3 of Olsen (1964a) (Fig. 3). Poor exposure prevented measurement of a new stratigraphic section, development of a magnetic polarity stratigraphy, or recovery of additional fossils. During previous studies, fossil vertebrates were recovered primarily from a 1-m interval near the base of the outcrop, overlain by approximately 4 m of clay and sand included in the Torreya Formation (Olsen, 1964a; Hunter and Huddlestun, 1982; Huddlestun, 1988). Hunter and Huddlestun (1982) included this outcrop in their *Pododesmus* biozone (see below).

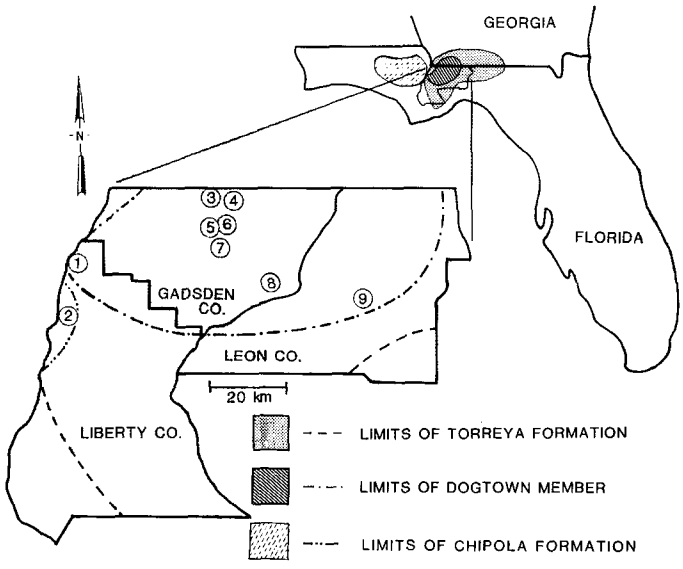


Figure 1. Index map of localities and areal distribution of Torreya Formation, Dogtown Member, and Chipola Formation in Florida and Georgia. Localities: 1, Rock Bluff; 2, Alum Bluff; 3, Milwhite Gunn Farm Mine; 4, Engelhard La Camelia Mine; 5, Floridin Corry Mine; 6, Floridin Smith Mine; 7, Quincy, location of the Quincy site; 8, Midway; 9, Tallahassee, location of the Seaboard and Tallahassee Water Works sites.

Alum Bluff Group

The Alum Bluff section is one of the most widely known and described outcrops in the southeastern United States (Schmidt, 1984; Johnson, 1989). More than 30 m of section is exposed, including ~10 m of the Alum Bluff Group (Johnson, 1989) (Fig. 4). The Chipola Formation is exposed at the base of the section and consists of as much as 4 m (at low river level) of extremely fossiliferous, sandy limestone (Schmidt, 1984; Johnson, 1989). The Chipola Formation represents a nearshore shelf environment with very little brackish influence (Vokes, 1989). The mollusc samples for $^{87}Sr/^{86}Sr$ analyses were collected from near the top of the Chipola (Fig. 4); no terrestrial vertebrate fossils were recovered from the Chipola Formation. Approximately 5 m of clayey, quartz sand with a discontinuous basal conglomerate overlies the Chipola Formation; this unit contains rare vertebrate fossils (see below) (Fig. 4). This sand unit has been referred to various formations, most recently the "Fort Preston Formation" (Puri and Vernon, 1964), the "Hawthorn Formation" (Schmidt,

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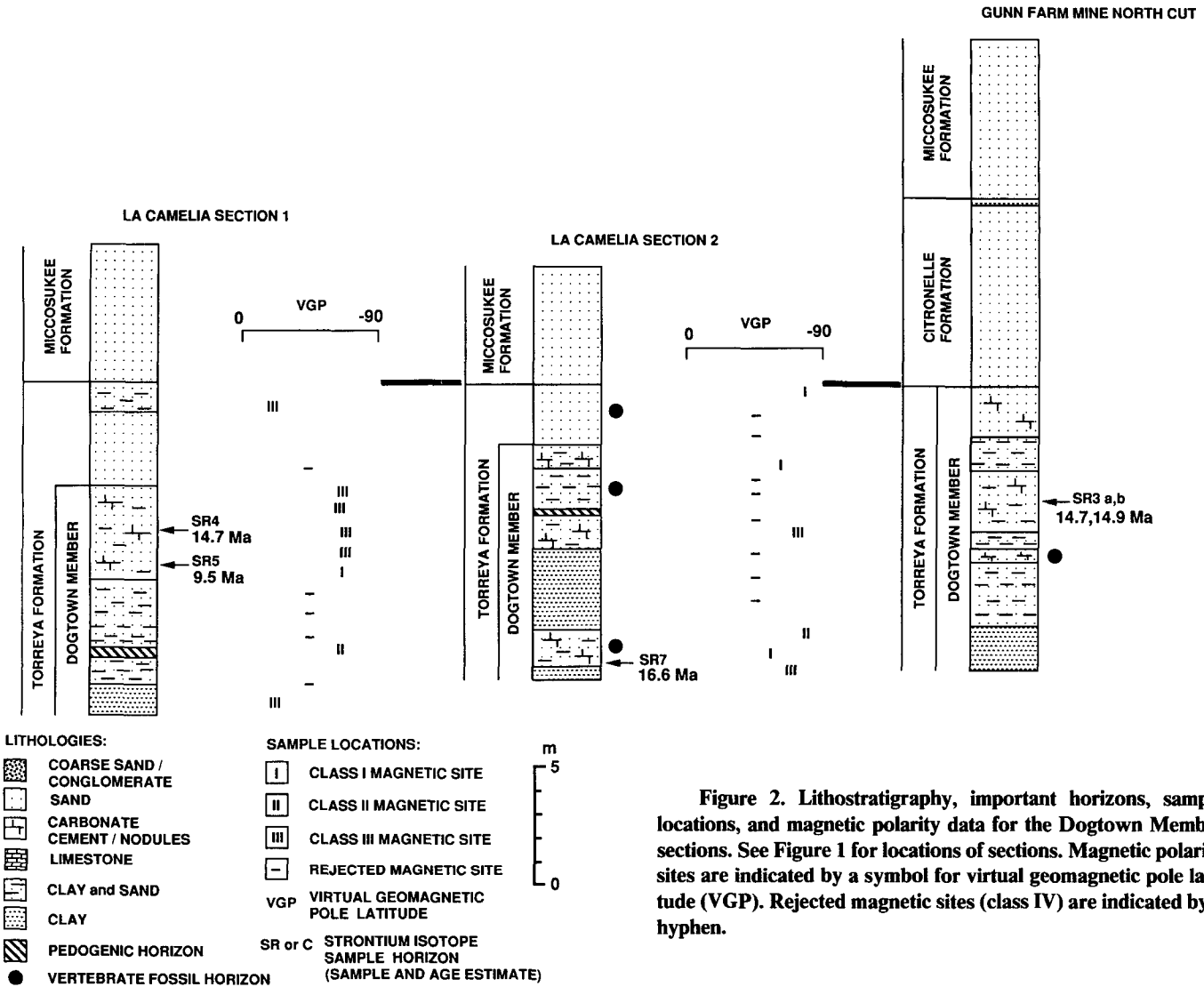


Figure 2. Lithostratigraphy, important horizons, sample locations, and magnetic polarity data for the Dogtown Member sections. See Figure 1 for locations of sections. Magnetic polarity sites are indicated by a symbol for virtual geomagnetic pole latitude (VGP). Rejected magnetic sites (class IV) are indicated by a hyphen.

1984), and the "Alum Bluff Group undifferentiated" (Johnson, 1989). Scott (1988) did not include this sand in the Hawthorn Group. For the purposes of this study, the clayey sand unit is referred to as the "Alum Bluff Group undifferentiated" after Johnson (1989); for brevity and distinction from the Chipola Formation, this sand unit will further be referred to as the "Fort Preston sand" after Puri and Vernon (1964), with the recognition that this designation is informal by definition (North American Commission on Stratigraphic Nomenclature, 1983) and that the unit may later be reassigned. Most authors (for example, Puri and Vernon, 1956, 1964; Schmidt, 1984) interpret an unconformity separating the "Fort Preston sand" from the underlying Chipola Formation; as a result, the section was not sampled for magnetic polarity studies.

INVERTEBRATE BIOCHRONOLOGY AND CORRELATIONS

The invertebrate paleontology of deposits referred to as the "Torreya and Chipola Formations" is prominent in discussions of stratigraphy and paleoecology of Miocene deposits in Florida. Gardner (1926) published a monograph of fossil molluscs from the Alum Bluff Group (which at that time encompassed most of the Hawthorn Group of Scott, 1988, and Huddlestun, 1988). Gardner (1926) considered the Chipola Formation (including the Dogtown Member of the Torreya Formation) to be "lower" Miocene in age. Puri and Vernon (1956) and Olson (1966) identified several species of molluscs, benthonic Foraminifera, and ostracodes found in fuller's earth clay mines south of the town of Quincy (Fig. 1); many of these species are also found in the Chipola Formation. On the basis of the overall faunal similarities with the Chipola Formation, Olson (1966) also assigned the fossiliferous sediments exposed in the fuller's earth mines near Quincy to the Chipola Formation, although other workers continued to assign the fuller's earth beds to the Hawthorn Formation, on the basis of lithology (for example, Patterson, 1974).

Banks and Hunter (1973, p. 355) originally proposed the Torreya Formation, based on faunal content, for "strata containing a macrofauna including *Pododesmus scopelus* Dall and a microfauna with *Miogypsina globulina* Michelotti," exposed at Rock Bluff (Fig. 1). At that time, the Dogtown Member had not been assigned to the Torreya Formation. Banks and Hunter (1973) indirectly correlated the Torreya Formation with very late *Catapsydrax dissimilis* Zone or more probably *C. stainforthi* Zone (planktonic foraminiferal zones upper N5 and N6 of Blow, 1969),

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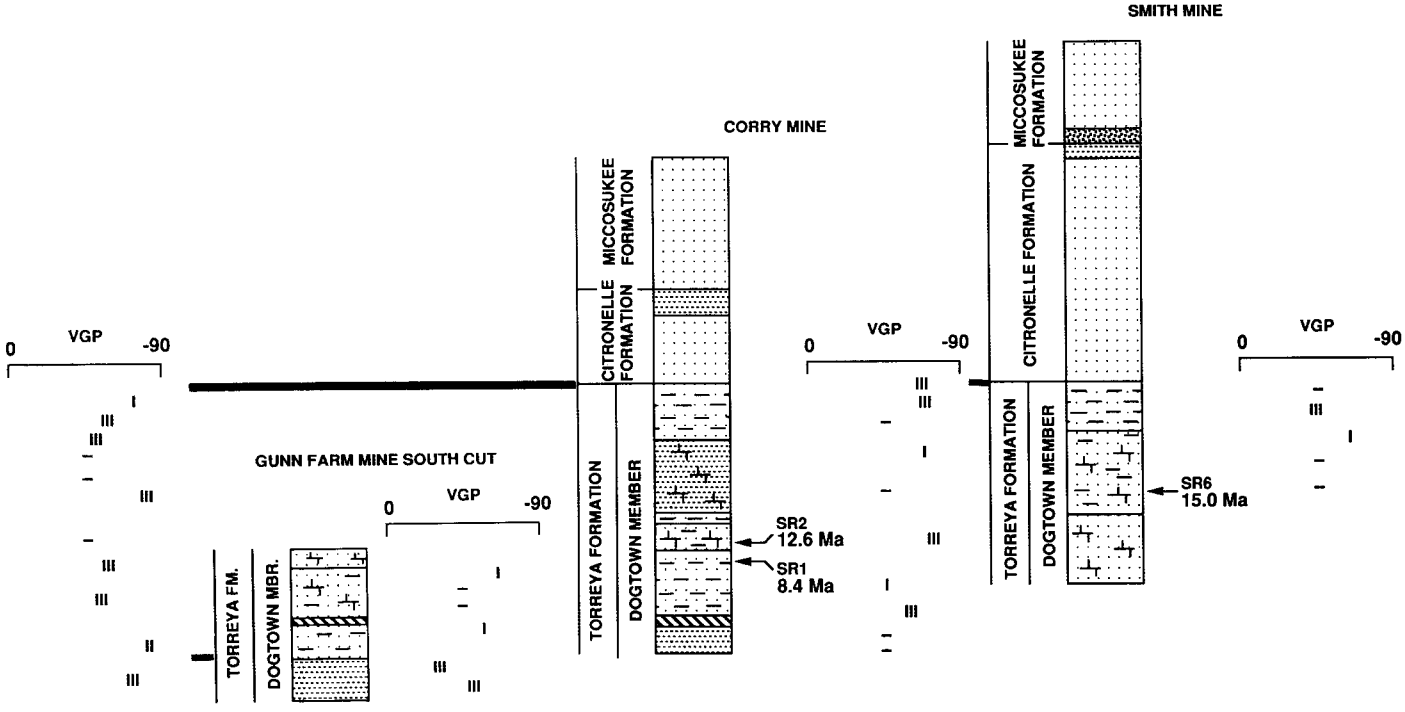


Figure 2. (*Continued*).

based on the range of the benthonic Foraminifera *Miogypsina* in Florida and the Gulf Coastal Plain. This was supported by the occurrence of early Hemingfordian land mammals (the Seaboard local fauna) in the lower Torreya Formation. These biochronologic data supported an early Miocene age for the Torreya Formation, older than that interpreted for the Chipola Formation (see below). Hunter and Huddlestun (1982) supported a similar pre-Chipola stratigraphic position and age for the Dogtown Member, based on the presence of late Hemingfordian mammals in the Dogtown Member (the Midway local fauna), despite the lack of diagnostic marine microfossils in the Dogtown or a known stratigraphic contact between the Chipola Formation and the Dogtown Member.

Hunter and Huddlestun (1982) established three molluscan biozones for their revised Torreya Formation, which included the Dogtown Member. These biozones generally correspond with lithostratigraphic units

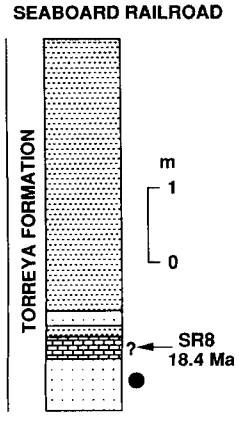


Figure 3. Lithostratigraphy, important horizons, and sample locations for the Seaboard section (after Olsen, 1964a). See Figure 2 for explanation of symbols.

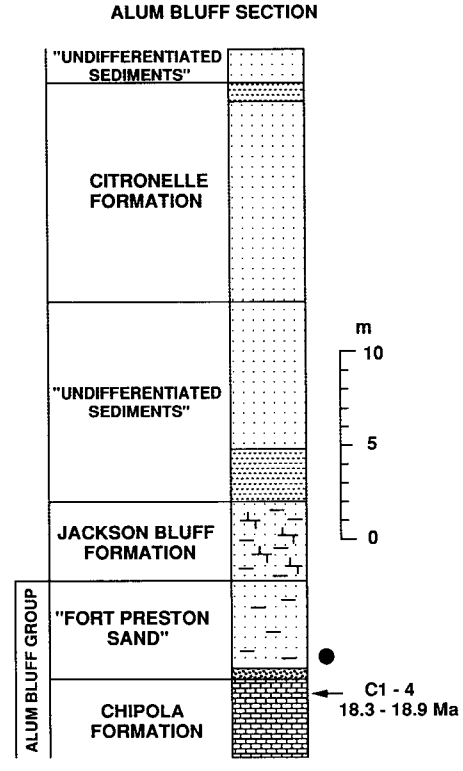


Figure 4. Lithostratigraphy, important horizons, and sample locations for the Alum Bluff section (after Johnson, 1989). See Figure 2 for explanation of symbols.

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and are useful in establishing the stratigraphic position of fossil vertebrate faunas from the Torreya Formation. The biozones are, in ascending order, the *Pododesmus scopelus* Assemblage Zone, the *Carolia floridana* Local Range Zone, and the *Chlamys nematopleura* Local Assemblage Zone (Fig. 5). The *Pododesmus* biozone includes the Torreya Formation *sensu* Banks and Hunter (1973), or the lower Torreya Formation stratigraphically below the Dogtown Member. This biozone contains the *Miogypsina* correlated by Banks and Hunter (1973) with planktonic foraminiferal zones upper N5 and N6. The *Carolia* and *Chlamys* biozones both occur within the Dogtown Member. The *Carolia* biozone occurs in a fossiliferous, sandy unit separating the fuller's earth clay beds, where two such beds are present, as at the La Camelia Mine (Fig. 3). The *Chlamys* biozone occurs above the fuller's earth clay beds. Neither the *Carolia* or *Chlamys* biozone contains diagnostic invertebrate microfossils, but were indirectly correlated with Zone N6 or early N7 based on the presence of late Hemingfordian land mammals in the Midway local fauna (Hunter and Huddlestun, 1982; Tedford and Hunter, 1984; Huddlestun, 1988).

The Chipola Formation has a long history of paleontological investigations, primarily for the rich molluscan fauna (~440 species) it contains (Gardner, 1926; Vokes, 1989, and references therein). Banks and Hunter (1973) noted similarities between the Torreya and Chipola invertebrate faunas and listed several molluscs in common to both formations; however, they determined that the Torreya Formation was stratigraphically lower (and older), based upon the superposition of the Chipola Formation over an erosional remnant of the lower Torreya Formation in the subsurface at Alum Bluff, the only known locality where the Torreya and Chipola Formations occur together. A Burdigalian (early Miocene) age for the Chipola Formation has been accepted by most authors, although Vokes (1965) suggested that the Chipola Formation may be slightly younger, on the basis of correlations of muricine gastropods with European species. Several species of planktonic Foraminifera and calcareous nannofossils have been reported from the Chipola Formation (Akers, 1972). Unfortunately, the species identified by Akers were not recovered from the Alum Bluff locality and are temporally long-ranging species that indicate only that the age could range from early Miocene (N5) to late middle Miocene (N13 or late CN5a) (Blow, 1969; Barron and others, 1985; Bolli and Saunders, 1985). The absence of *Orbulina* was cited by Akers (1972) as negative evidence for a correlation no later than the N8/N9 boundary (the "*Orbulina* datum"). Additionally, the absence of *Miogypsina* in the Chipola Formation (Akers, 1972) was cited as negative evidence that the Chipola was younger than the Torreya Formation (Banks and Hunter, 1973), although *Miogypsina* survived as late as N12 in the Caribbean (Blow, 1969). Gibson (1967, p. 643) and Akers (1972) felt that the Chipola Formation correlated best with the *Globigerinatella insueta* Zone of Bolli (1957) and Blow (1959), equivalent to zones N7 through N8 of Blow (1969), but were not able to document the presence of the critical species.

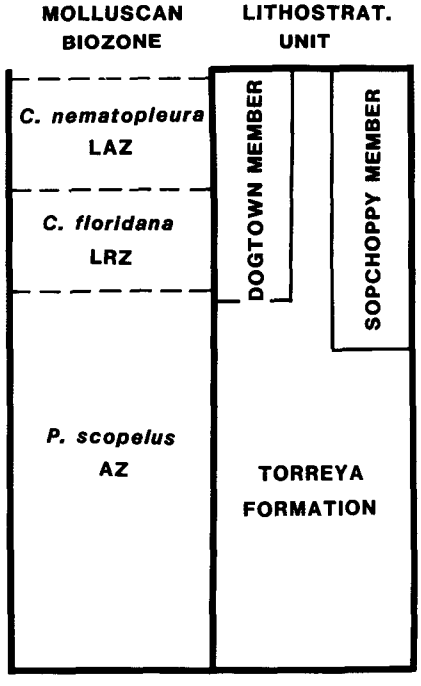


Figure 5. Molluscan biostratigraphy of the Torreya Formation, after Hunter and Huddlestun (1982). The Dogtown Member includes the *Carolia* and *Chlamys* biozones; the *Pododesmus* biozone is restricted to the lower Torreya Formation.

VERTEBRATE BIOCHRONOLOGY

Vertebrate faunas are distributed throughout the Torreya Formation (Fig. 6). Two local faunas are known from the lower Torreya Formation (below the Dogtown Member), the Seaboard and Griscom Plantation local faunas. The Seaboard local fauna (l.f.) (Fig. 1) is the richest vertebrate locality within the lower Torreya Formation. Taxa present include the parahippine horse *"Parahippus" leonensis*, the anchitherine horses *Anchitherium clarencei* and *Archaeohippus blackbergi*, the protoceratid *Prosynthetoceras texanus* and other large artiodactyls, and the heteromyid rodent *Proheteromys floridanus* (Olsen, 1964a). Tedford and Hunter (1984) assigned an early Hemingfordian land-mammal age to the Seaboard l.f. The Griscom Plantation l.f. is not as large as the Seaboard l.f., but also contains *"Parahippus" leonensis* and is considered early Hemingfordian in age (Sellards, 1916; Simpson, 1930, 1932a; Hulbert and MacFadden, 1991). In addition to the Griscom Plantation and Seaboard l.f.'s, the Chattahoochee Bluff locality in the lower Torreya Formation has

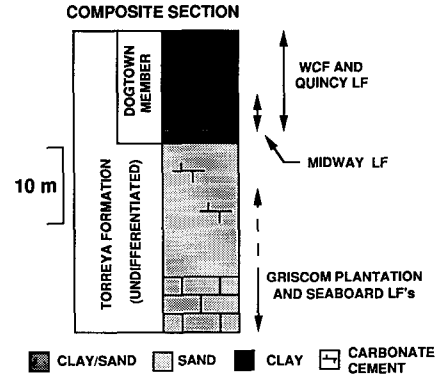


Figure 6. Composite stratigraphic section showing distribution of important Torreya Formation vertebrate faunas. The Willacoochee Creek Fauna (WCF) is early Barstovian; Quincy l.f. is late Hemingfordian or early Barstovian; Midway l.f. is late Hemingfordian; Griscom Plantation and Seaboard l.f.'s are early Hemingfordian. The Willacoochee Creek Fauna and Midway and Quincy l.f.'s occur at about the same lithostratigraphic horizon but are of different land-mammals ages.

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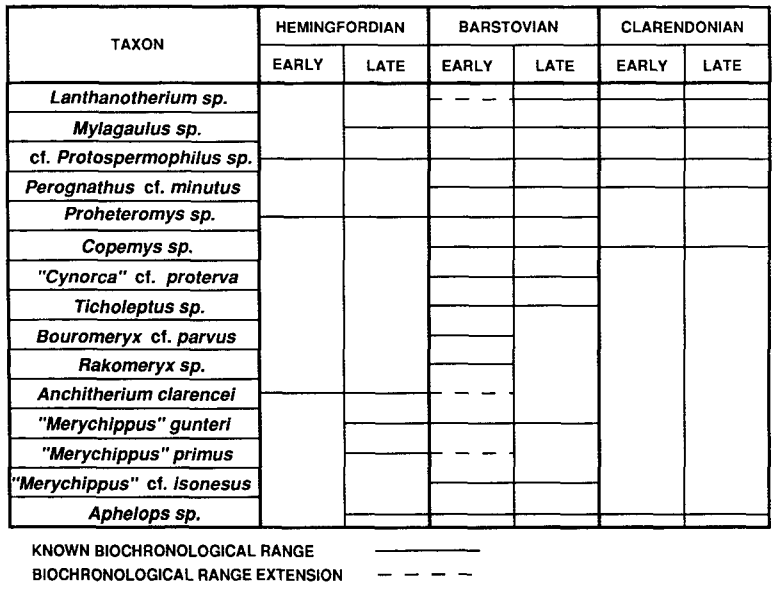


Figure 7. Vertebrate biochronology of the Willacoochee Creek Fauna. An early Barstovian land-mammal age is indicated by the overlapping ranges of several mammals.

produced an isolated molar that is probably referable to *"Parahippus" leonensis* (Olsen, 1968).

Four faunas are known from the Dogtown Member: the Midway, Quincy, Tallahassee Water Works, and Willacoochee Creek Faunas, representing two different stages of land-mammal evolution. The Midway l.f., recovered near the town of Midway in southeastern Gadsden County (Fig. 1), was collected from the Dogtown Member above the upper fuller's earth bed within the *Chlamys* biozone (Simpson, 1930; Hunter and Huddlestun, 1982). Taxa present include the merychippine horse *"Merychippus" gunteri*, the anchitherine horses *Anchitherium clarencei* and *Archaeohippus blackbergi*, the dromomerycid artiodactyl *Aletomeryx*, a primitive racoon (near *Arctonasua* or *Cyonasua*), *Proheteromys*, and the mylagaulid rodent *Mesogaulus* (Sellards, 1916; Simpson, 1930, 1932a; Wood, 1932; Tedford and Hunter, 1984). Tedford and Hunter (1984) assigned a late Hemingfordian age to the Midway l.f.

The Quincy l.f. was recovered from between two fuller's earth beds at the base of the local section (the *Carolia* biozone), and from the overlying sand and clay above the upper fuller's earth beds (the *Chlamys* biozone) (Sellards, 1916; Simpson, 1930, 1932a, 1932b). The terrestrial fauna consists of only a few teeth and postcranial elements referred to *"Merychippus" gunteri* (Hulbert and MacFadden, 1991). Tedford and Hunter (1984) united the Quincy and Midway l.f.'s as the Midway Fauna based on the occurrence of *"M." gunteri* at both localities, similar lithostratigraphic position, and presumably similar age. Similarities in lithostratigraphic position and the common occurrence of *"M." gunteri* at the Midway, Quincy, and Willacoochee Creek Faunas (Fig. 6), but significant differences in age, do not support the unification of the Midway and Quincy l.f.'s into one fauna. The Tallahassee Water Works l.f. (Fig. 1; exact location unknown) is also very small, consisting of several associated lower molars of the rhinocerotid *Aphelops* sp. and postcranial material of a small artiodactyl (Colbert, 1932). Placement of the Tallahassee Water Works locality in either the lower Torreya Formation or the Dogtown Member is not clear, based on the section described by Colbert, although he did mention the association of clay beds and *Carolia floridana* with the vertebrate fossils; it is therefore probable that the Tallahassee Water Works l.f. was collected from the Dogtown Member. The age of the Tallahassee Water Works l.f. is also equivocal, but is no older than late Hemingfordian, on the basis of the range of *Aphelops* (Tedford and others, 1987).

The Willacoochee Creek Fauna was collected from the four fuller's earth mines included in this study (Fig. 1). The Willacoochee Creek Fauna is the largest vertebrate fauna known from the Torreya Formation, consisting of at least 68 vertebrate taxa including 29 mammals (Bryant, 1991). Fossils were recovered from three different stratigraphic levels within the Dogtown Member, representing the *Carolia* and *Chlamys* biozones of Hunter and Huddlestun (1982) (Fig. 3). An early Barstovian land-mammal age is indicated by the presence or stage of evolution of several biochronologically useful taxa, including the heteromyid rodent *Perognathus*, the mylagaulid rodent *Mylagaulus*, three merychippine horses, and the artiodactyls *Rakomeryx*, *Bouromeryx* cf. *B. parvus*, *Ticholeptus*, and *"Cynorca"* cf. *"C." proterva* (Fig. 7).

No terrestrial mammals are yet known from the Chipola Formation, but a small vertebrate fauna has been recovered from the "Fort Preston sand" directly above the Chipola Formation at Alum Bluff (Fig. 4). Olsen (1964b, 1968) reported a partial horse mandible with two teeth, which he referred to *"Parahippus" leonensis*, recovered from the "Choctawhatchie Stage" (= "Fort Preston sand" of this study) directly above the Chipola Formation. Re-evaluation of this specimen reveals that it actually represents a small anchitherine horse. Additional terrestrial vertebrates were recovered from the same stratigraphic unit during the present study, including a very small rhinocerotid, a protoceratid, and an equid mandible with two teeth representing *"Merychippus" gunteri*. Despite extensive screen-washing efforts, terrestrial microfossils have not been recovered from the "Fort Preston sand." Nevertheless, the co-occurrence of these four macrovertebrate taxa suggests a late early or early middle Miocene age, probably late Hemingfordian or early Barstovian and correlative with the Midway or Willacoochee Creek l.f.'s of the Dogtown Member.

STRONTIUM ISOTOPE CHRONOLOGY

Field and Laboratory Methods

Eight biogenic carbonate samples were collected from seven stratigraphic horizons of the Dogtown Member, one sample from the lower Torreya Formation at the Seaboard locality, and four samples from the Chipola Formation exposed at Alum Bluff (Figs. 2–4). Samples from the Torreya Formation are shells of the marine molluscs *Ostrea* sp. or *Cras-*

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sostrea sp., *Chlamys nematopleura*, and *Carolia floridana*. Samples from the Chipola Formation are all *Mercenaria langdoni*. The mineralogy of the shells of modern representatives of the genera *Ostrea*, *Crassostrea*, and *Chlamys* is known to be calcite (Milliman, 1974). *Carolia* is an extinct genus of the family Anomiidae; *Anomia*, a modern representative of that family, has a shell which is 83%–95% calcite, the remainder aragonite (Milliman, 1974). Modern representatives of *Mercenaria* have an aragonitic shell (Milliman, 1974).

To assess the degree of preservation, thin-section and X-ray diffraction analyses of samples SR3b, SR7, and SR8 (Torreya Formation) were completed and indicated a calcite composition with no obvious signs of diagenesis. X-ray diffraction patterns of all four samples from the Chipola Formation indicate an aragonite composition. In addition, the entire collection of 13 samples was further analyzed for diagenesis using Sr/Ca ratios (Brand and Veizer, 1980; Veizer, 1983). Powder was abraded from each shell and dissolved in 10 ml of 0.5N HCl, filtered, and diluted with 1N HCl. All measurements were performed on a Perkin Elmer 3100 atomic absorption spectrometer.

Analytical procedures for Sr isotopic analyses are exactly the same as those outlined in Webb and others (1989), with the exception of the Chipola samples, which were mounted on Re ribbon for ionization rather than Ta ribbon used for the Torreya samples. Shell samples were carefully cleaned, and powder was abraded from the interior of the shells to obtain the most pristine carbonate and avoid surface impurities. The samples were dissolved and subjected to standard ion-exchange column techniques. Isotopic ratios were measured on a VG Isomass 354 mass spectrometer. All measurements are normalized to $^{86}Sr/^{88}Sr = 0.11940$, measurements of NBS standard $SrCO_3$ (NBS-987) = 0.710244, and long-term analytical precision (2σ) is estimated at 1.5×10^{-5}.

To estimate ages for the measured $^{87}Sr/^{86}Sr$ ratios, we utilized the Miocene sea-water evolution curves of Hodell and others (1991). These curves were chosen because of the high data density (and, consequently, lower errors) relative to other compilations for this time period (Richter and DePaolo, 1988; Miller and others, 1991). Age estimates based on all these compilations (adjusted for differences in laboratory standards), however, are presented in Table 1. It should be noted that error calculations for the individual regressions (time segments) are variable and do not include errors associated with the analytical error applicable to individual sample values (1.5×10^{-5}). This error is relatively small compared to the more common 95% confidence-interval errors of ±0.5–2.3 m.y. associated with the scatter of these particular reference data. These errors are not additive. Hodell and others (1991) cited errors of ±0.74 for the 24.0–16.0 Ma interval and ±1.36 for the 16.0–8.0 Ma interval. Because analytical errors are small and regression errors vary along the length of the regressed line (progressively increasing away from the midpoint), we conservatively estimate our correlations to the Hodell and others (1991) curve to be approximately ±1.0 m.y. for the 24.0–16.0 Ma interval and approximately ±1.5 m.y. for the 16.0–8.0 Ma interval. Although somewhat arbitrary, these error estimates are large relative to the reproducibility of the samples and should be reasonable estimates. Miller and others (1991) cited stratigraphic resolutions (based on sample reproducibility and regression equation slope) of ±0.5 m.y. (25.1–14.6 Ma interval) and ±2.3 m.y. (14.5–8.3 Ma interval).

To estimate ages for the measured $^{87}Sr/^{86}Sr$ ratios using the sea-water evolution curve of Richter and DePaolo (1988), a regression was calculated for $^{87}Sr/^{86}Sr$ versus time for all data in the interval from 18.0 to 13.5 Ma. Following the methods of McKenzie and others (1988), age was regressed as the dependent variable against $^{87}Sr/^{86}Sr$, which yields an equation of

$$\text{age (Ma)} = 8868.69 - 12491.04 \times (^{87}Sr/^{86}Sr).$$

This equation has a slope of 79.5×10^{-6}/m.y., and a correlation coefficient (r^2) of 0.68. A 95% confidence interval of ±1.4 m.y. was calculated from an analysis of variance about the regression line; this error is at the midpoint of the regression boundaries and is the minimum error associated with this regression. Two samples, SR1 and SR5, have anomalously high $^{87}Sr/^{86}Sr$ ratios and fall outside the boundaries of this regression (see below); age estimates for these two samples were calculated by two-point extrapolation between successive data points on the Richter and DePaolo (1988) curve.

TABLE 1. TAXA ANALYZED, Sr/Ca DATA, $^{87}Sr/^{86}Sr$ RATIOS, AND REGRESSION AGE ESTIMATES OF MOLLUSCS FROM THE TORREYA AND CHIPOLA FORMATIONS

Sample	Taxa	Sr/Ca × 10^{-3}	$^{87}Sr/^{86}Sr$	Model ages		
				(1)	(2)	(3)
SR1	O,C	1.07	0.708906	8.4	8.1	9.1
SR2	O,C	1.26	0.708808	12.6	14.9	15.0
SR3a	Ch	2.31	0.708758	14.7	15.6	15.8
SR3b	Ch	2.67	0.708753	14.9	15.6	15.9
SR4	O,C	1.02	0.708759	14.7	15.6	15.8
SR5	O,C	0.507	0.708882	9.5	9.0	11.0
SR6	Ch	0.723	0.708751	15.0	15.7	15.9
SR7	Ca	1.61	0.708701	16.6	16.3	16.8
SR8	O,C	0.474	0.708590	18.4	17.7	18.6
C1	M	4.02	0.708593	18.4	17.6	18.6
C2	M	4.24	0.708600	18.3	17.5	18.5
C3	M	4.09	0.708563	18.9	18.0	19.1
C4	M	3.74	0.708580	18.6	17.8	18.8

Note: NBS-987 is measured at 0.710244 ± 15 (2σ). $^{86}Sr/^{88}Sr$ is normalized to 0.11940. Samples SR1–SR7 are from the Dogtown Member of the Torreya Formation (Fig. 2). SR8 is from the Seaboard locality, lower Torreya Formation (Fig. 3), and samples C1–C4 are from the Chipola Formation (Fig. 4). Taxa: O,C, *Ostrea* or *Crassostrea*; Ch, *Chlamys nematopleura*; Ca, *Carolia floridana*; M, *Mercenaria langdoni*. Sr/Ca data are expressed as molar ratios. Error associated with $^{87}Sr/^{86}Sr$ measurements is $\pm 1.5 \times 10^{-5}$ (2σ). Model ages (1) Hodell and others (1991); estimated errors are ±1.0 m.y. (24.0–16.0 Ma) and ±1.5 m.y. (16.0–8.0 Ma). (2) Richter and DePaolo curve; calculated minimum error is ±1.4 m.y. (see text). (3) Miller and others (1991); errors are ±0.5 m.y. (25.1–14.6 Ma) and ±2.3 m.y. (14.6–8.3 Ma).

Results

The taxa analyzed, Sr/Ca ratios, $^{87}Sr/^{86}Sr$ ratios, and age estimates are reported in Table 1 (modified from MacFadden and others, 1991). The Sr/Ca ratios are within the broad ranges reported for marine carbonates (Milliman, 1974; Veizer, 1983), although samples SR5, SR6, and SR8 from the Torreya Formation are at the extreme low end of the range of modern calcitic molluscs (Milliman, 1974). These data indicate that a diagenetic overprint may have affected the strontium composition of these three samples. If so, the ratios for SR5, SR6, and SR8 are likely to yield minimum age estimates if the samples interacted with downward-percolating ground water that would carry younger (higher $^{87}Sr/^{86}Sr$) strontium.

Age estimates for the Torreya Formation range from 8.4 ± 1.5 to 16.6 ± 1.0 Ma for eight samples from the Dogtown Member and 18.4 ± 1.0 Ma for one sample from the Seaboard locality (Table 1). Samples SR1 and SR5 from the Torreya Formation have anomalously high $^{87}Sr/^{86}Sr$ ratios, yielding age estimates of 8.4 and 9.5 (±1.5) Ma. The age estimates for samples SR1 and SR5 are inconsistent with all other geochronologic evidence and are overlain by samples with more concordant ages (Fig. 3), and are interpreted to have not retained a primary marine signal. Sample SR8 from the Seaboard locality also has a very low Sr/Ca ratio but an age estimate consistent with the vertebrate and invertebrate biochronology. The 18.4 ± 1.0 Ma age estimate for sample SR8 is therefore interpreted as a minimum age, because the diagenetic overprint exhibited in samples SR1 and SR5 appears to favor unchanged or increased $^{87}Sr/^{86}Sr$ ratios and

younger age estimates. Two samples from the same horizon (SR3a and SR3b) yield identical ratios (within analytical precision) and age. The age estimate for sample SR2 is 12.6 ± 1.5 Ma using the Hodell and others (1991) model, but 14.9 ± 1.4 Ma with the Richter and DePaolo curve and 15.8 ± 0.5 Ma with the Miller and others (1991) model. This is an artifact of the statistical models used (see below). The four samples from the Chipola Formation yield remarkably consistent Sr/Ca ratios, $^{87}Sr/^{86}Sr$ ratios, and age estimates of 18.3–18.9 (±1.0) Ma. The Sr/Ca ratios of the Chipola samples are high and within the range of modern marine aragonitic molluscs (Milliman, 1974). These samples are considered to have retained a primary marine signal.

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Although the ages discussed above are obtained from regression equations, the assessment of errors, the root of true chronostratigraphic resolution, is more complex. For unknown samples, factors such as analytical error, diagenetic alteration, and the slope of the sea-water curve are obviously important. The slope of the sea-water curve is determined by the rather arbitrary selection of linear, quadratic, or higher functions for segments of the curve. In addition, the sea-water evolution curve itself also has inherent uncertainties associated with the analytical accuracy and precision, diagenetic alteration, and the method used to assign time (or ages) to the reference data sets (magnetostratigraphy, biostratigraphy, sedimentation rate, and so on). As in other geochronologic studies, confidence accrues from concordance of results utilizing different systems. In this case, the data and regressions of Richter and DePaolo (1988) and Miller and others (1991) were also used to make age estimates for these samples. As can be seen in Table 1, agreement is generally good. Contrasting results (for example, sample SR2) reflect both the differences in reference data and the statistical strategies used; the limits of the regression equation used by Miller and others (1991), Hodell and others (1991), and our equation for the data of Richter and DePaolo (1988) are different. In addition, Miller and others did not treat age as the dependent variable. Taken together, however, these age estimates reflect the current state of Sr-isotope chronostratigraphy for this part of the Miocene. Although the curve is relatively steep in this time period, uncertainties related to statistical approaches and reference data remain. Improved resolution will primarily result from better-constrained reference sections, rather than improved analytical techniques.

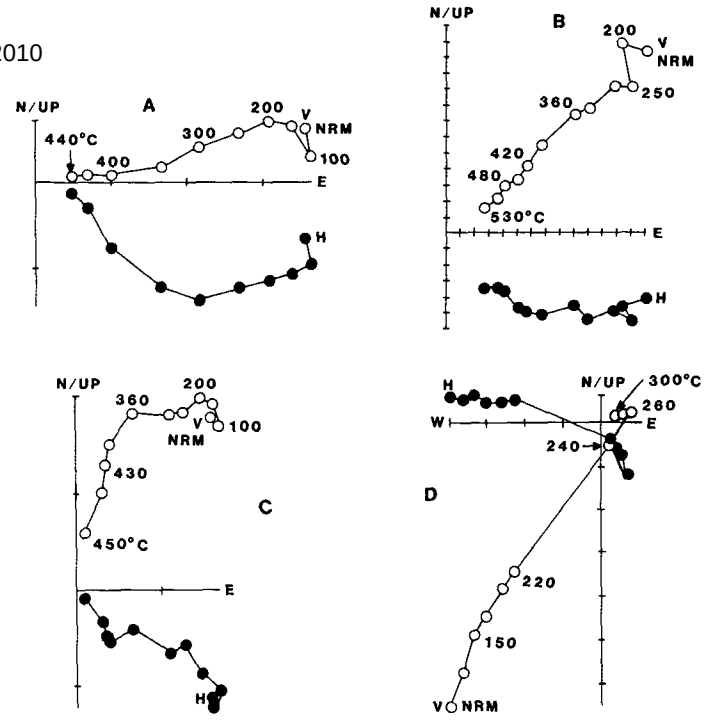


Figure 8. Representative vector demagnetization diagrams of thermally demagnetized samples from the Dogtown Member. Secondary overprints are removed below 260 °C, generally isolating a characteristic remanence direction above 400 °C. Alternating field (AF) demagnetization was generally ineffective. Open circles represent the vertical component of the demagnetization vector; filled circles are the horizontal component.

MAGNETOSTRATIGRAPHY

Methods and Results

Three separately oriented samples were collected from each of 60 individual sample horizons (sites) from the Dogtown Member exposed in fuller's earth mines in northern Gadsden County. Sites were sampled at ~1-m intervals and placed in detailed measured sections (Fig. 3). Samples were cut into 2.5-cm cubes and measured at the Paleomagnetics Laboratory at the University of Florida. Measurements were performed in a shielded room in which the Earth's ambient field is reduced to ~200 gammas (Scott and Frohlich, 1985). Procedures were performed using an ScT cryogenic magnetometer and Schonstedt AF and thermal demagnetization equipment.

For the Milwhite north cut, La Camelia section 1, Corry, and Smith Mines, one sample from each site was subjected to AF demagnetization (10 to 60 mT) and one to thermal demagnetization (100 to 630 °C). The demagnetization behavior of each site was then evaluated; thermal demagnetization was interpreted as most effective in isolating characteristic remanence directions. In some cases, secondary overprints were removed below 260 °C (Fig. 8), generally isolating characteristic remanence directions between 400 and 550 °C. Above these temperatures, decay directions generally became unstable and ambiguous. On the basis of these results, the third sample from each site was also subjected to thermal demagnetization. The Milwhite south cut and La Camelia section 2 sections were measured subsequent to analyses of the other four sections; all three samples from each site at the Milwhite south cut and La Camelia section 2 sections were therefore subjected to thermal demagnetization.

Typical isothermal remanence magnetization (IRM) curves for several samples have steep initial slopes in applied fields to as much as 150 mT, and generally reach saturation in peak fields of less than 250 mT (Fig. 9). In some cases, saturation was not reached in fields to as high as 350 mT, although IRM increased rapidly in fields to as high as 150 mT. These results, along with stable demagnetization behavior above 400 °C, indicate that magnetite or titanomagnetite is probably the primary magnetic mineral, with a small contribution from a second, higher-coercivity mineral. The slight upward drift in some IRM samples and the low-temperature overprint could possibly result from the presence of hematite or iron sulfides in the sediments. In their petrologic examination of the Dogtown Member, Weaver and Beck (1977) reported the presence of pyrite, which may oxidize to hematite (Channell and others, 1982).

As a result of demagnetization, the polarity of 34 of the 60 sites is considered unambiguous and interpretable, either as class I (N = 10), class II (N = 3), or class III (N = 21) sites (*sensu* Opdyke and others, 1977). Characteristic directions were determined for each sample, using stable end points, and Fisher site and formation means were calculated for class I sites (Table 2). The formation mean direction (dec. = 182.2°, inc. = –37.0°,

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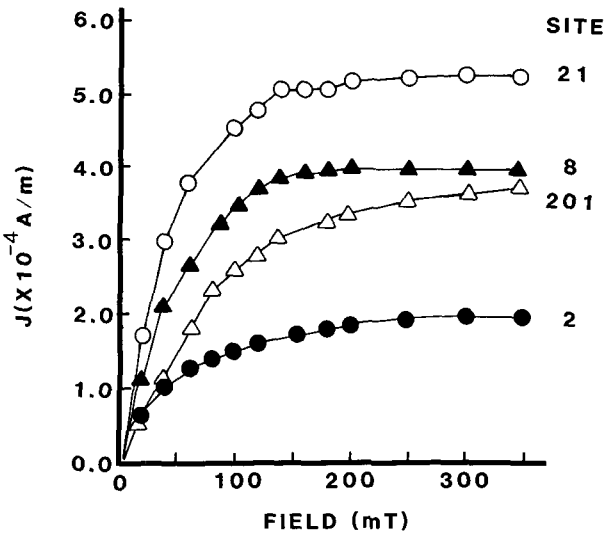


Figure 9. Representative isothermal remanence magnetization (IRM) curves for several samples. All samples have initially steep slopes in applied fields to as much as 150 mT and generally saturate by 250 mT. Some samples do not saturate to as much as 350 mT, indicating the presence of a second, higher-coercivity mineral.

TABLE 2. PALEOMAGNETIC SITE-MEAN DIRECTIONS AND FISHER STATISTICS FOR CLASS I SITES FROM THE DOGTOWN MEMBER

Site	Dec.	Inc.	N	k	α_{95}	R
			Engelhard La Camelia section 1			
66	181.3	-24.0	3	29.0	23.3	2.9
			La Camelia section 2			
211	234.1	-55.3	3	12.2	37.0	2.8
220	146.2	-48.9	3	5.7	57.0	2.7
223	186.9	-33.1	3	11.9	37.4	2.8
			Milwhite Gunn Farm Mine north cut			
12	174.3	-32.1	3	8.3	45.7	2.8
			Gunn Farm Mine south cut			
202	217.6	-24.5	3	4.8	64.1	2.6
205	186.5	-16.7	3	12.6	36.4	2.8
			Floridin Corry Mine			
43	139.5	-26.0	3	27.3	24.1	2.9
46	158.3	-23.5	3	10.3	40.6	2.8
			Floridin Smith Mine			
22	148.7	-53.3	3	13.6	34.9	2.9
			Composite site mean			
	182.2	-37.0	10	8.8	17.2	9.0

N = 10, k = 8.8, α_{95} = 17.2°, R = 9.0) is statistically significant (Tarling, 1983). These results are certainly less than those necessary to interpret a complex magnetostratigraphy. All of our sites, however, are of reversed polarity, class I sites have a significant formation mean direction, interpretable sites are stratigraphically well distributed throughout the entire section, and there are no major unconformities. Additionally, fossil vertebrates collected from different stratigraphic levels in the studied sections record no apparent difference in their stage of evolution, indicating that they were deposited in a relatively short period of time. It therefore seems probable that the Dogtown Member exposed in the study area was deposited within a single chron.

Correlation with the Geomagnetic Polarity Time Scale

The following data constrain the correlation of the magnetic polarity stratigraphy from the Dogtown Member with the geomagnetic polarity time scale of Berggren and others (1985).

(1) A diagnostic early Barstovian vertebrate fauna (the Willacoochee Creek Fauna) has been collected from the sampled sections. The Hemingfordian-Barstovian boundary is placed within Chron C5B-R at approximately 15.9 Ma, and the early Barstovian–late Barstovian boundary within Chron C5AD at about 14.8 Ma (Burbank and Barnosky, 1990; MacFadden and others, 1990; Woodburne and others, 1990).

(2) The occurrence of early Hemingfordian mammals (the Seaboard l.f.) in the lower Torreya Formation stratigraphically below the Dogtown Member provides a lower constraint of about 18 Ma (Tedford and others, 1987).

(3) The occurrence of *Miogypsina globulina* in the lower Torreya Formation provides a lower constraint of planktonic foraminiferal zones upper N5 and N6 (Hunter and Huddlestun, 1982). Barron and others (1985) assigned ages of about 19 to 17.6 Ma for zones upper N5 and N6, and Berggren and others (1985) correlated those zones with Chrons C6 through C5D.

(4) $^{87}Sr/^{86}Sr$ age estimates for the Dogtown Member in the study area are between 14.7 ± 1.5 and 16.6 ± 1.0 Ma, and 18.4 ± 1.0 Ma for the underlying Torreya Formation exposed at the Seaboard locality.

(5) The entire Dogtown Member in the study area is of reversed polarity with no apparent, long-term hiatuses.

Given the constraints above, the Dogtown Member in the northern Gadsden County area correlates best with Chron C5B-R (Fig. 10). Alternative correlations are possible with either the short reversed polarity interval within Chron C5B-N or with Chron C5AD-R, but these are unlikely because unreasonably high sedimentation rates would be necessary to deposit the Dogtown Member within either one of these chrons, and the high density of sample horizons would most likely detect all polarity intervals present. Furthermore, correlation with Chron C5B-R is

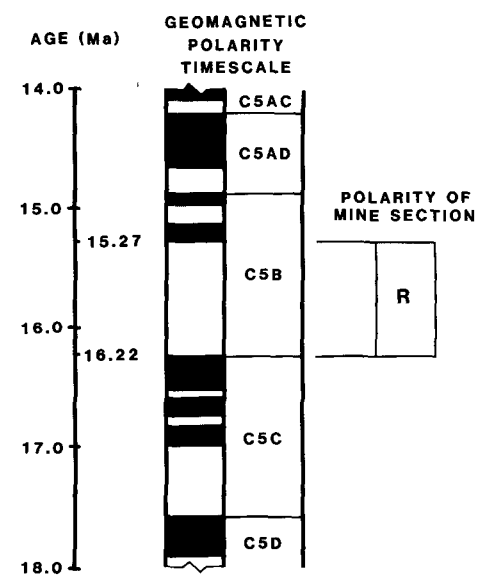


Figure 10. Correlation with the geomagnetic polarity time scale of Berggren and others (1985). The most likely correlation is with Chron C5B-R, based on associated geochronologic data.

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more consistent with the fossil vertebrate biochronology and $^{87}Sr/^{86}Sr$ age estimates. Berggren and others (1985) assigned boundaries of 15.27 and 16.22 Ma to Chron C5B-R.

DISCUSSION

Age of the Torreya Formation

Hunter and Huddlestun (1982) estimated the age of the Torreya Formation as no older than about 20.5 Ma to as young as about 17 Ma, based on what was then known of the vertebrate and invertebrate biochronology. New data presented in this paper refine that age estimate, particularly for the upper boundary of the Dogtown Member and the Torreya Formation (Fig. 11). Integration of biochronology, strontium-isotope chronostratigraphy, and magnetic polarity stratigraphy restricts the age of the Dogtown Member in the northern Gadsden County area to between 15.3 and 15.9 Ma. The minimum age of about 15.3 Ma is provided by correlation of the reversed polarity of the sediments with Chron C5B-R; a maximum age of about 15.9 Ma is supported by the presence of early Barstovian land mammals and the beginning of the Barstovian land-mammal age (MacFadden and others, 1990). This age bracket is further supported by $^{87}Sr/^{86}Sr$ age estimates of 14.7 ± 1.5 and 16.6 ± 1.0 Ma for the best-preserved fossil molluscs in the section. The presence of late Hemingfordian land mammals in the Dogtown Member from the Midway l.f. suggests that the Dogtown Member is time transgressive; however, this locality is no longer accessible for study. A $^{87}Sr/^{86}Sr$ age estimate of 18.4 ± 1.0 Ma from the Seaboard locality in the lower Torreya Formation is consistent with the occurrence of an early Hemingfordian land-mammal fauna at that locality and supports indirect correlations with planktonic foraminiferal zones upper N5 and N6 for the lower Torreya. Thus, the Torreya Formation is late early Miocene to early middle Miocene in age, between about 19 and 15.3 Ma.

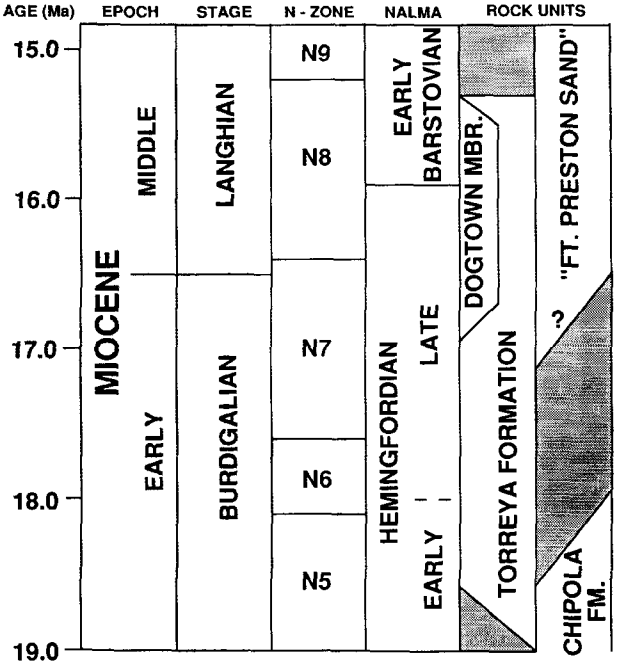


Figure 11. Summary of geochronology, showing temporal relationships between the Torreya and Chipola Formations, and the "Ft. Preston sand." Stippled areas are unrepresented time intervals. Abbreviations: N-ZONE, planktonic foraminiferal zonation; NALMA, North American land-mammal age. Epoch and stage boundary ages after Berggren and others (1985); planktonic foraminiferal zonation boundaries after Barron and others (1985); North American land-mammal age boundaries after Tedford and others (1987) and MacFadden and others (1990).

Age of the Chipola Formation

The Chipola Formation has been generally correlated with planktonic foraminiferal zones N7 and N8, transitional between the early and middle Miocene, but younger than the Torreya Formation. This interpretation has been supported by several He-U dates between 14.1 and 17.8 Ma and averaging 16.1 ± 1.0 Ma (Bender, 1973). New $^{87}Sr/^{86}Sr$ age estimates for the Chipola Formation are older than previous interpretations (Fig. 11). Four mollusc samples collected near the top of the Chipola Formation yield $^{87}Sr/^{86}Sr$ age estimates between 18.3 and 18.9 (±1.0) Ma, which are older than the age interpreted for the Chipola Formation by previous authors. Furthermore, the occurrence of a small vertebrate fauna of late Hemingfordian or early Barstovian age in the "Fort Preston sand," overlying the Chipola Formation at Alum Bluff, supports the new age estimates presented here.

Previous interpretations of the age of the Chipola Formation have relied upon correlations with planktonic foraminiferal zones based on negative evidence by Gibson (1967) and Akers (1972). A correlation based on positive evidence is restricted only to between zones N5 (early Miocene) and N13 (late middle Miocene). The discrepancy between the $^{87}Sr/^{86}Sr$ age estimates and the invertebrate biochronology is probably due to the indirect correlations and reliance upon negative evidence. Given the wide range of He-U ages, it is likely that some or all of these samples have lost He, which would generate an age estimate which is too young. A diagenetic overprint of the Sr age estimates is not likely because the molluscs are still of aragonitic composition, Sr/Ca ratios are consistent with modern aragonitic molluscs, and the vertebrate fauna from the overlying "Fort Preston sand" provides an independent confirmation of the ages. Furthermore, as the general diagenetic trend appears to be toward unchanged or higher ratios (and therefore younger ages), if the Chipola Formation molluscs have been seriously affected by diagenesis, the age estimates would represent minimum ages. Thus, an age of about 18.3 Ma for the top of the Chipola Formation is not contradicted by available evidence.

Conclusions

Strata assigned to the Chipola and Torreya Formations are known to occur together only in the subsurface at Alum Bluff, where the Chipola overlies the Torreya (Scott, 1988; Huddlestun, 1988). At outcrops where the Torreya and Chipola Formations were sampled for this study, however, there is considerable chronologic overlap between the lower Torreya Formation and upper Chipola Formation (Fig. 11). These new chronologic data indicate that the Torreya and Chipola Formations probably interdigitate in this area, that the Chipola Formation is largely older than the Torreya Formation, and that the Dogtown Member of the Torreya Formation is definitely younger than the Chipola Formation, based upon representative $^{87}Sr/^{86}Sr$ signatures. In addition, the new data and interpretations presented here support the view that the Alum Bluff and Hawthorn Groups are relatively contemporaneous and may, in part, represent correlative, marginal-marine and offshore-shelfal facies of late early and early middle Miocene age. $^{87}Sr/^{86}Sr$ chronostratigraphy is a valuable tool for

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interpreting the age of biogenic carbonates preserved in nearshore marine deposits and will be useful for deciphering the chronostratigraphy of coastal-plain deposits and integrating vertebrate and invertebrate biochronology.

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